

Markov Chain Cubature for Bayesian Inference

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DataSig

A rough path between
mathematics and data science



The
**Alan Turing
Institute**

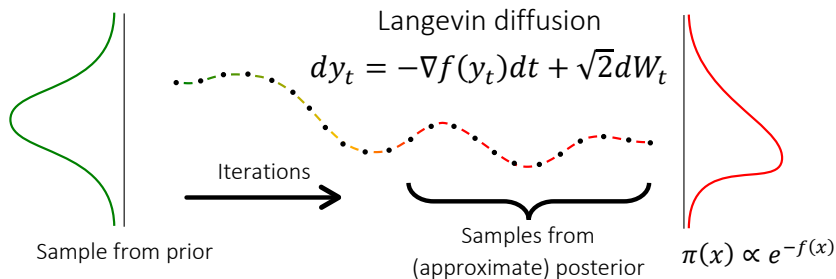


Outline

- ① Markov Chain Monte Carlo
- ② One-step cubature formula
- ③ Distribution compression
- ④ Error analysis of cubature
- ⑤ Numerical experiments
- ⑥ Conclusion and ongoing work
- ⑦ References

Markov Chain Monte Carlo

A fundamental challenge in Bayesian inference is computing integrals with respect to posterior distributions and Markov Chain Monte Carlo (MCMC) is widely regarded as the “go-to” approach for these problems.



Under mild conditions on $f : \mathbb{R}^d \rightarrow \mathbb{R}$, the Langevin SDE admits a unique strong solution that is ergodic with stationary measure $\pi(x) \propto e^{-f(x)}$ [1].

Unadjusted Langevin Algorithm (ULA):

$$Y_{n+1} := Y_n - \nabla f(Y_n)h + \sqrt{2}W_n$$

Convergence of Langevin Monte Carlo

For sufficiently small h , ULA “ends up close” to the target distribution [2]. However, ULA is a Markov chain and might not explore the space quickly.

Broadly speaking, this leads to two possible approaches

- Run N independent ULA chains and sample points at a fixed time T .
- Run a ULA chain over a long time horizon and sample at each step.

The latter strategy is much preferred by practitioners, but relies on the Markov chain to have a **fast mixing time**.

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Motivating questions

Can Langevin dynamics be simulated as a cloud of (dependent) points?
Accuracy? Computational cost? Exploration of parameter space?

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One-step cubature formula

Recall the Langevin diffusion with invariant measure $\pi \propto e^{-f}$ is given by

$$dy_t = -\nabla f(y_t) dt + \sqrt{2} dW_t, \quad (1)$$

By Taylor expanding (1), we can see that y has the following moments:

$$\mathbb{E}[(y_t - y_s) | y_s] = -\nabla f(y_s) h + O(h^2), \quad (2)$$

$$\mathbb{E}[(y_t - y_s)^{\otimes 2} | y_s] = 2hI_d + O(h^2), \quad (3)$$

for $s, t \geq 0$, where $h = t - s > 0$.

Construction of a first order one-step cubature rule started at y_s

We want to construct a cloud of points and weights $\{(x_i, w_i)\}_{1 \leq i \leq N}$ with

$$\mu := \sum_{i=1}^N w_i x_i = -\nabla f(y_s) h, \quad \Sigma := \sum_{i=1}^N w_i (x_i - \mu)(x_i - \mu)^\top = 2hI_d.$$

One-step cubature formula via Hadamard matrices

The Hadamard (or Walsh) matrices are defined inductively as

$$H_1 := (1), \quad H_{2^{k+1}} := \begin{pmatrix} H_{2^k} & H_{2^k} \\ H_{2^k} & -H_{2^k} \end{pmatrix}.$$

Let $n := 2^{\lceil \log d \rceil}$ and define n vectors $\{e_i\}_{1 \leq i \leq n}$ in $\mathbb{R}^d$ as columns of H_n

$$H_n = \begin{pmatrix} e_1 & e_2 & \cdots & e_{n-1} & e_n \\ (n-d) \times n \text{ matrix} \end{pmatrix},$$

and another vectors $\{e_i\}_{n+1 \leq i \leq 2n}$ as $e_i := -e_{i-n}$ for $n+1 \leq i \leq 2n$.

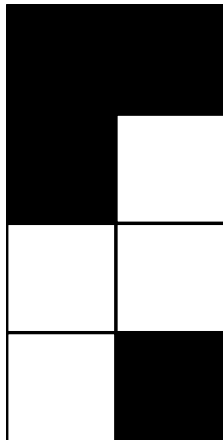
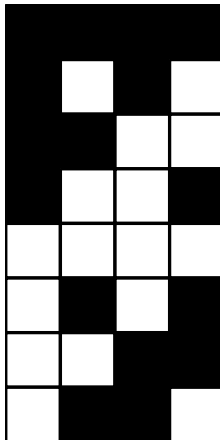
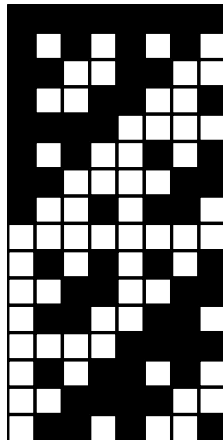
Theorem (Victoir (2005))

Let $X \sim \text{Uniform}(\{e_i\}_{1 \leq i \leq 2n})$. Then X is a symmetric random vector with covariance matrix:

$$\mathbb{E}[X^{\otimes 2}] = I_d.$$

Thus, X matches the first 3 moments of the standard normal distribution.

One-step cubature formula via Hadamard matrices

 H_2  H_4  H_8 

One-step cubature formula via Hadamard matrices

Therefore, in each step, we define $2^{\lceil \log d \rceil + 1}$ new cubature particles as

$$y_i^{\text{new}} := y_s - \nabla f(y_s)h + \sqrt{2h}e_i, \quad (4)$$

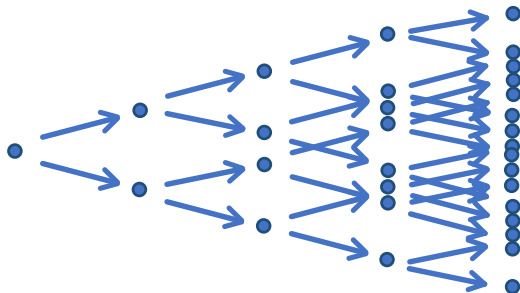
each with equal weight. The moments of (4) will then have $O(h^2)$ error. Equation (4) is a particular example of “Cubature on Weiner Space” [4]. However, even in the one-dimensional setting, there is a clear problem!

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After each step, the number of particles increases by a factor of $\approx 2d$.

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Thus, for SDE cubature to be practical, we require an algorithm that can “compress” (or reduce the support of) discrete probability distributions. In addition to just resampling particles, there are a variety of algorithms where the compressed measure accurately integrates certain functions.

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Test functions are explicitly specified by the user (e.g. polynomials)

- High order Recombination (Litterer and Lyons (2012), Tchernychova and Lyons (2016), Cosentino et al. (2020))

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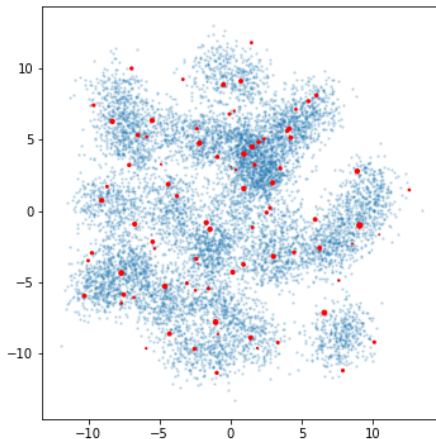
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Test functions come from a reproducing kernel (i.e. MMD distance)

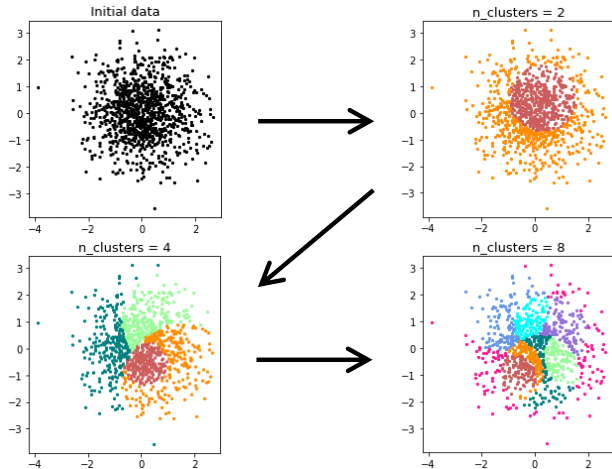
- Kernel Herding (Chen et al. (2010))
- Kernel Thinning (Dwivedi and Mackey, (2021))
- Kernel Recombination (Hayakawa et al. (2022, 2023))
- Stein Thinning (Riabiz et al. (2022))

Distribution compression



Example of Stein Thinning [14], which is designed to thin MCMC samples

Applying distribution compression locally



A standard ball tree algorithm for partitioning points is implemented in

```
from sklearn.neighbors import BallTree
```

A basic cubature algorithm

Step 0. Generate a cloud of N points Y_0 with uniform weights $w_0 = \frac{1}{N}$ from a prior distribution.

Step 1. In each step, generate a new cloud of $2^{1+\lceil \log d \rceil} N$ points by

$$Y_{n+1,i} := Y_n - \nabla f(Y_n)h + \sqrt{2h}e_i,$$

with weights $w_{n+1,i} := 2^{-(1+\lceil \log d \rceil)} w_n$.

The vectors $\{e_i\}$ come from the “Hadamard” cubature formula.

Step 2. Put the points into N smaller “patches” via a ball tree algorithm.

Step 3. On each patch, resample a point (or reduce small clouds of points/weights using a distribution compression algorithm).

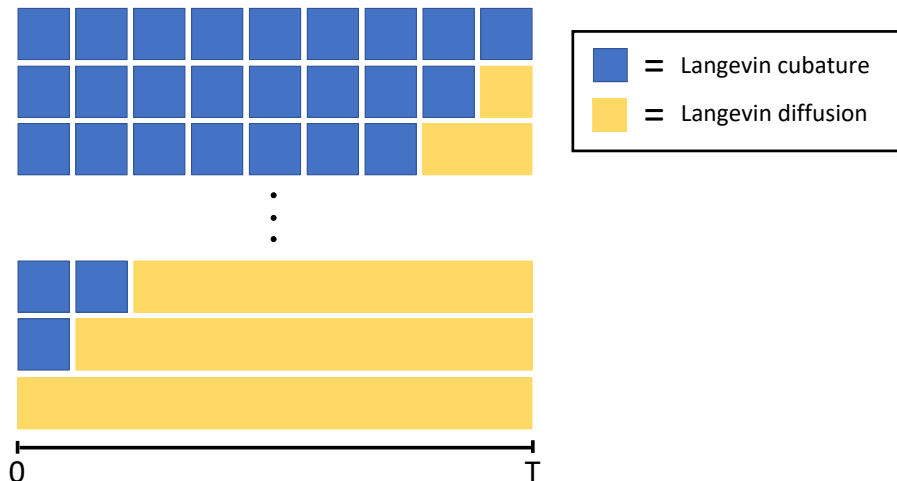
Step 4. Repeat steps 1 – 3.

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Error analysis of cubature

We want to compare n steps of Langevin cubature to the SDE at $T = nh$.



Error analysis of cubature

For a smooth test function $F : \mathbb{R}^d \rightarrow \mathbb{R}$, let

$$P_t F : y \mapsto \mathbb{E}[F(y_t) | y_0 = y],$$

$$Q_k F : Y \mapsto \mathbb{E}[F(Y_k) | Y_0 = Y],$$

be the semigroups associated with the Langevin diffusion and cubature (Y_k is ULA with Hadamard random vectors). By a telescoping sum trick,

$$\begin{aligned} P_T F - Q_n F &= \sum_{k=0}^{n-1} Q_{n-(k+1)} (P_{t_{k+1}} F) - Q_{n-k} (P_{t_k} F) \\ &= \sum_{k=0}^{n-1} Q_{n-(k+1)} (P_h - Q_1) (P_{t_k} F), \end{aligned}$$

where the cubature algorithm uses a step size of $h > 0$ and $t_k := kh$.

Error analysis of cubature

Since Q_1 is a Markov operator, it follows from [15, Theorem 13.2] that

$$\begin{aligned}\|P_T F - Q_n F\|_{L_\mu^2} &\leq \sum_{k=0}^{n-1} \|Q_{n-(k+1)}(P_h - Q_1)(P_{t_k} F)\|_{L_\mu^2} \\ &= \sum_{k=0}^{n-1} \left\| \underbrace{Q_1 \cdots Q_1}_{\substack{n-(k+1) \\ \text{times}}} (P_h - Q_1)(P_{t_k} F) \right\|_{L_\mu^2} \\ &\leq \sum_{k=0}^{n-1} \|(P_h - Q_1)(P_{t_k} F)\|_{L_\mu^2},\end{aligned}$$

where μ is the prior distribution (that is, $y_0 \sim \mu$) and $\|\cdot\|_{L_\mu^2}$ is the norm

$$\|G\|_{L_\mu^2} := \left(\int_{\mathbb{R}^d} \|G(x)\|^2 d\mu(x) \right)^{\frac{1}{2}}.$$

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Hence, there are two different operators to estimate: $(P_h - Q_1)$ and P_{t_k} .

Error analysis of cubature (based on Talay-Tubaro [16])

Using the notation $F_t := P_t F$, we can Taylor expand $(P_h - Q_1)(P_t F)$ since

$$\mathbb{E}[F_t(y_h) | y_0 = y] = F_t(y) + (\Delta F_t(y) - \nabla F_t(y) \nabla f(y))h + R_1(t, y)h^2,$$

$$\mathbb{E}[F_t(Y_1) | Y_0 = y] = F_t(y) + (\Delta F_t(y) - \nabla F_t(y) \nabla f(y))h + R_2(t, y)h^2,$$

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$$\begin{aligned} \mathbb{E}_{y_0 \sim \mu} \left[\left\| (P_T F - Q_n F)(y_0) \right\|^2 \right]^{\frac{1}{2}} &\leq \sum_{k=0}^{n-1} \mathbb{E}_{y_0 \sim \mu} \left[(C(1 + \|y_0\|^m)e^{-\alpha t_k})^2 \right]^{\frac{1}{2}} h^2 \\ &\leq \sum_{k=0}^{n-1} \tilde{C} h^2 e^{-\alpha t_k} \\ &\leq \tilde{C} \left(\frac{1}{\alpha} + h \right) h. \end{aligned}$$

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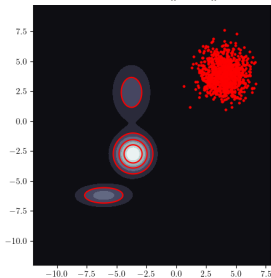
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Outline

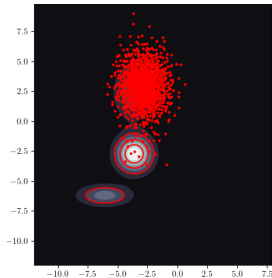
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Langevin cubature on a Gaussian mixture model

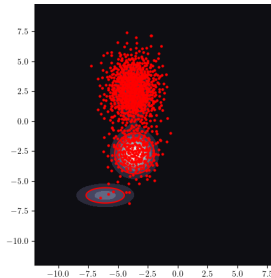
Iter 0; $\|\mu - \hat{\mu}\| = 10.036$; $\|\Sigma - \hat{\Sigma}\| = 10.019$



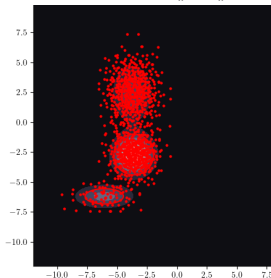
Iter 25; $\|\mu - \hat{\mu}\| = 4.971$; $\|\Sigma - \hat{\Sigma}\| = 8.144$



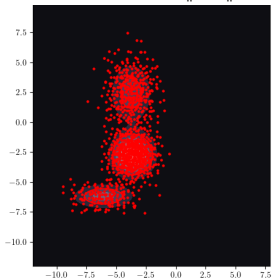
Iter 100; $\|\mu - \hat{\mu}\| = 3.288$; $\|\Sigma - \hat{\Sigma}\| = 3.767$



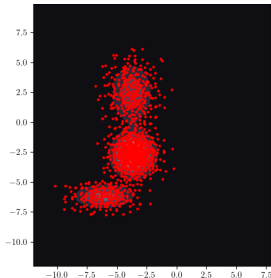
Iter 250; $\|\mu - \hat{\mu}\| = 1.590$; $\|\Sigma - \hat{\Sigma}\| = 0.640$



Iter 500; $\|\mu - \hat{\mu}\| = 0.488$; $\|\Sigma - \hat{\Sigma}\| = 0.311$



Iter 1000; $\|\mu - \hat{\mu}\| = 0.016$; $\|\Sigma - \hat{\Sigma}\| = 0.227$



Cubature vs MCMC on a Gaussian mixture model

This experiment was run using my (suboptimal) code in PyTorch
(i.e. [MCCube](#) and [BlackJAX](#) times will be different)

Algorithm	Time (s)	Iterations	$\ \mu - \hat{\mu}\ $ (3.d.p)
Langevin cubature	32.9	1000	0.016
Single MCMC chain (ULA)	362.5	1000000 (+ 1000 burn-in)	0.047

Cubature parameters:

- Step size, $h = 0.1$
- Number of particles, $N = 1024$

MCMC parameters:

- Step size, $h = 0.1$

Cubature vs Stein Variational Gradient Descent

Stein Variational Gradient Descent (SVGD), due to Liu & Wang (2016), is a very popular interacting particle algorithm for Bayesian inference.

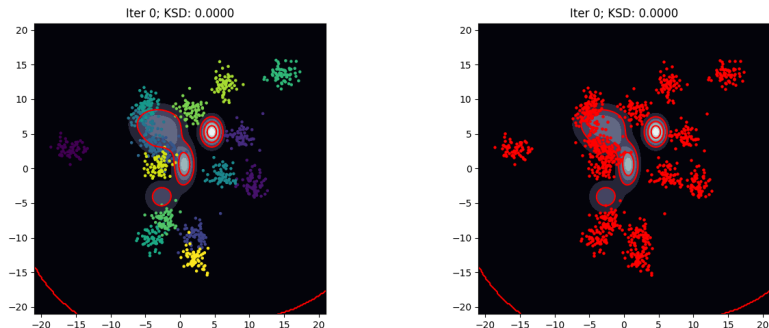
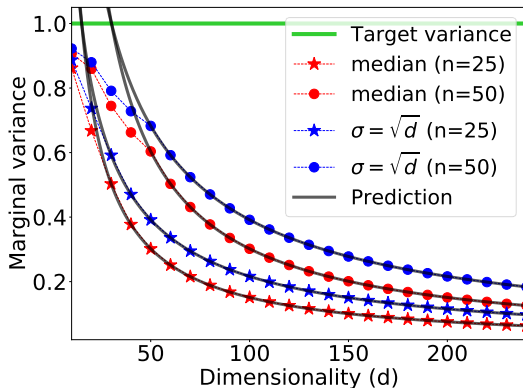


Figure: (left) Langevin Cubature. (right) Stein Variational Gradient Descent.

The variance collapse of SVGD

However, without enough particles, SVGD exhibits “variance collapse” and behaves similar to gradient-based single particle optimisation [19].



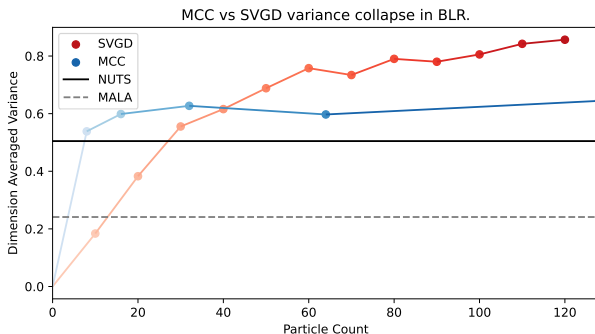
Furthermore, SVGD scales quadratically with the number of particles.

Cubature vs SVGD vs MCMC for logistic regression

In addition to multivariate Gaussian distributions, [19] also considers

$$f(\theta) := -\mathbf{y}^\top \mathbf{x} \theta + \sum_{i=1}^m \log(1 + \exp(-\theta^\top \mathbf{x}_i)) + \frac{\alpha}{2} \|\theta\|_2^2,$$

with input vectors $\mathbf{x} = \{\mathbf{x}_i\}_{i=1}^m \in \mathbb{R}^{m \times d}$, labels $\mathbf{y} = \{\mathbf{y}_i\}_{i=1}^m \in \{-1, 1\}^m$ and regression parameters $\theta \in \mathbb{R}^d$, where $d = 128$, $m = 500$ and $\alpha = 1$.



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- Langevin cubature can outperform MCMC on a 2D Gaussian mixture and also performs well on a (synthetic) Bayesian logistic regression.
- Unlike SVGD, it can avoid $O(N^2)$ complexity in the number of particles.

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Ongoing work

- GPU-capable software is available at github.com/tttc3/MCCube with documentation [here](#).

(currently being developed
by Thomas Coxon)

- Further numerical
experiments against
SVGD and MCMC



MCCube

Markov chain cubature via JAX

Thank you
for your attention!

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Error analysis of cubature (based on Talay-Tubaro [16])

Classical SDE Theory ([1, Prop 4.2], [17, Prop 3.1 and Thm 3.4])

Suppose f is a *confining potential* ($\lim_{|x| \rightarrow \infty} f(x) = \infty$ and $\int_{\mathbb{R}^d} e^{-\beta f(x)} dx < \infty$ for $\beta > 0$), ∇f is *smooth with bounded derivatives* and $\exists c > 0$ such that $\langle x, \nabla f(x) \rangle \geq c\|x\|^2$. Then

- For $m \geq 1$, there exists constants $C_m, \alpha_m > 0$ such that for all $y_0 \in \mathbb{R}^d$ and $t \geq 0$,

$$\mathbb{E}[\|y_t\|^m] \leq C_m(1 + \|y_0\|^m e^{-\alpha_m t}). \quad (5)$$

- Suppose $F \in C^\infty(\mathbb{R}^d)$ and its derivatives have at most polynomial growth at ∞ . Define $u(t, y) := \mathbb{E}[F(y_t) | y_0 = y]$. Then for any given multi-index $I = (i_1, \dots, i_k)$, there exists $C_I, \alpha_I > 0$ and a positive integer m_I such that for $y \in \mathbb{R}^d$ and $t \geq 0$,

$$\left| \frac{\partial^{|I|}}{\partial^{i_1} y^1 \dots \partial^{i_k} y^{i_k}} (u(t, y)) \right| \leq C_I (1 + \|y\|^{m_I}) e^{-\alpha_I t}. \quad (6)$$

In addition, there exists $C, \alpha > 0$ and $m \geq 1$ such that for all $y \in \mathbb{R}^d$ and $t \geq 0$,

$$\left| u(t, y) - \mathbb{E}_{x \sim \pi}[F(x)] \right| \leq C(1 + \|y\|^m) e^{-\alpha t}, \quad (7)$$

where $\pi(x) \propto e^{-f(x)}$ is the *stationary distribution* of the Langevin diffusion $\{y_t\}$.